

Chemical signaling in biofilm-mediated biofouling

Received: 25 January 2022

Accepted: 14 August 2024

Published online: 30 September 2024



Xiaobo Liu¹✉, Ling Zou¹, Boqiao Li¹, Patrick Di Martino², Daniel Rittschof³, Jin-Long Yang⁴, James Maki⁵, Weijie Liu⁶✉ & Ji-Dong Gu^{7,8}✉

Biofouling is the undesirable accumulation of living organisms and their metabolites on submerged surfaces. Biofouling begins with adhesion of biomacromolecules and/or microorganisms and can lead to the subsequent formation of biofilms that are predominantly regulated by chemical signals, such as cyclic dinucleotides and quorum-sensing molecules. Biofilms typically release chemical cues that recruit or repel other invertebrate larvae and algal spores. As such, harnessing the biochemical mechanisms involved is a promising avenue for controlling biofouling. Here, we discuss how chemical signaling affects biofilm formation and dispersion in model species. We also examine how this translates to marine biofouling. Both inductive and inhibitory effects of chemical cues from biofilms on macrofouling are also discussed. Finally, we outline promising mitigation strategies by targeting chemical signaling to foster biofilm dispersion or inhibit biofouling.

Biofouling is the accumulation of both microorganisms and macroorganisms as well as their metabolites on submerged surfaces in aquatic environments. Biofouling is a global problem associated with huge economic losses across many sectors, especially for maritime industries, energy production and militaries¹. The global cost of biofilms has been estimated to account for almost US\$4,000 billion per year, with around two-thirds attributed to corrosion². Thus, there is an increasing demand for developing anti-biofouling strategies to mitigate structural failures and financial losses.

In any aquatic ecosystems, biofouling on a physical surface includes both microfouling and macrofouling (Fig. 1). As the initiation of biofouling, microfouling is the formation of a slimy conditioning film and subsequently a microbial biofilm^{3,4}. Importantly, biofilm development is usually regulated by chemical signaling molecules, such as cyclic dinucleotides (for example, cyclic (c)-di-guanosine monophosphate (GMP))^{5,6}, indole⁷ and quorum-sensing (QS) molecules^{3,8}. Furthermore,

biofilm on a surface is highly dynamic in terms of community structures, microenvironments, metabolisms and functions⁹. In marine environments, such bioactive features generate chemical cues that in turn may regulate the settlement of microscopic propagules of sessile macrofouling organisms (for example, invertebrates and macroalgae, in Fig. 1), the survival of which requires settlement of planktonic larvae or zoospores¹⁰. Selective settlement of these larvae or spores is highly regulated by chemical cues from the biofilms on surfaces^{3,11,12}, and thus, in many circumstances, biofilms mediate biofouling.

The fact that biofilm development is highly regulated by chemical signaling molecules and that biofilm-derived chemical cues may induce or inhibit the settlement of larvae and zoospores support biofouling control with chemical cues. Therefore, revisiting the roles of these signals and cues from the viewpoint of chemical biology provides a promising avenue for anti-biofouling. In this Perspective, we briefly review the biofouling processes and the roles of signaling molecules in biofilm

¹School of Environmental and Biological Engineering, Nanjing University of Science and Technology, Nanjing, China. ²Groupe Biofilm et Comportement Microbien aux Interfaces, Laboratoire ERRMECe, Cergy Paris Université, Cergy-Pontoise, France. ³Duke Marine Laboratory, Nicholas School of the Environment, Duke University, Beaufort, NC, USA. ⁴International Research Center for Marine Biosciences, Ministry of Science and Technology, Shanghai Ocean University, Shanghai, China. ⁵Department of Biological Sciences, Marquette University, Milwaukee, WI, USA. ⁶Jiangsu Key Laboratory of Phylogenomics and Comparative Genomics, School of Life Sciences, Jiangsu Normal University, Xuzhou, China. ⁷Environmental Engineering Program, Guangdong Technion-Israel Institute of Technology, Shantou, China. ⁸Guangdong Provincial Key Laboratory of Materials and Technologies for Energy Conversion, Guangdong Technion-Israel Institute of Technology, Shantou, China. ✉e-mail: xbliu@njust.edu.cn; leonliu2013@126.com; jjdong.gu@gtit.edu.cn

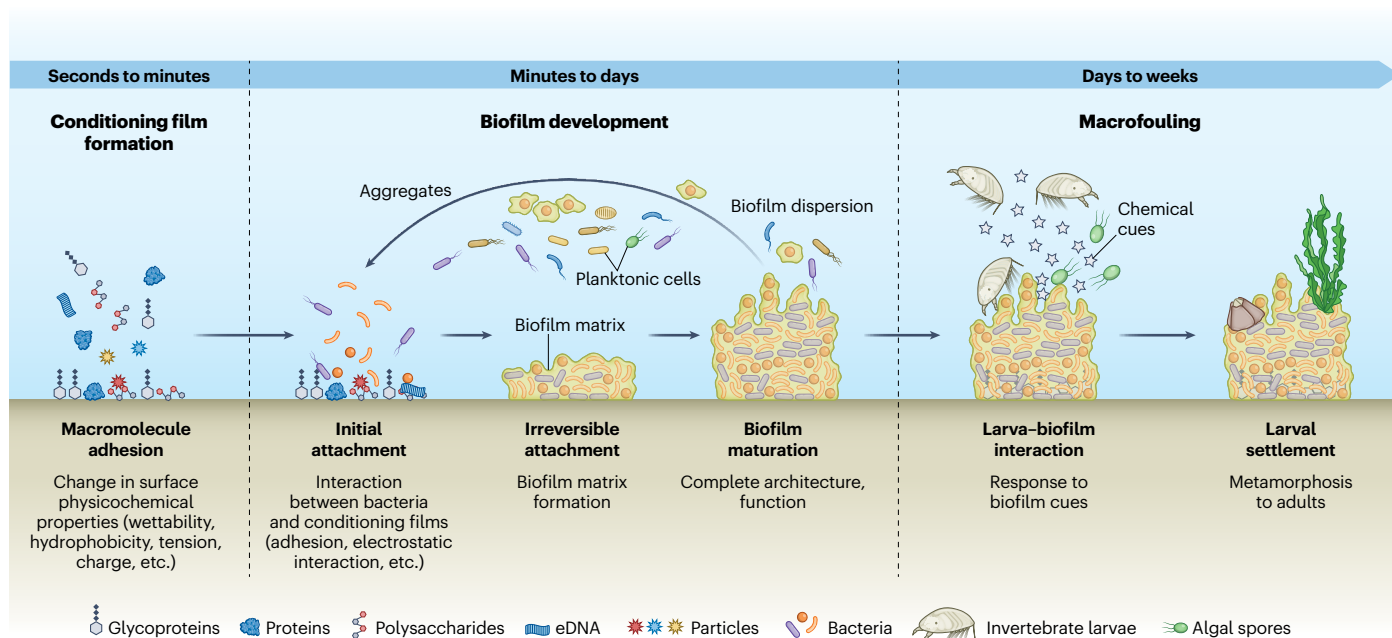


Fig. 1 | Overview of the processes of biofouling in natural environments. General stages of biofouling at a freshly immersed surface. Natural biofilms in aquatic or marine environments are a multispecies community predominated by bacteria. Fouling of macroorganisms is highly affected by the preformed biofilms due to their ubiquitous distribution in aquatic habitats. Referring to the

bacterial biofilm life cycle, this multispecies process therefore is hypothesized with four stages: (1) initial cell attachment to a conditioning film, (2) irreversible attachment, (3) maturation of the biofilm, and (4) biofilm dispersion at the final stage.

formation and dispersion before focusing on biofilm-derived chemical cues for macrofouling propagules. We argue that there are potential mitigation solutions by targeting chemical signaling that result in biofilm dispersion or impede biofouling. To conclude, we highlight that biofilms mediate much of biofouling and provide future perspectives on biofouling mitigation through chemical biology approaches.

Biofilm development of model species

Typically, biofilm development is characterized by initial formation of a conditioning film^{13,14}, the process of biofilm formation from initial cell attachment to biofilm maturation¹⁵ and biofilm dispersal^{16,17} (Fig. 1). Here we mainly catalog chemical signaling molecules that regulate the biofilm life cycle to identify biochemical targets for chemical biology investigations of new therapies for biofouling challenges.

Conditioning film formation

Conditioning film is formed immediately when a pristine surface is immersed in aqueous environments. This is a consequence of rapid adsorption of surface-active organic compounds (for example, humic substances, amino acids, polysaccharides, proteins and other metabolites) and inorganic compounds from the water column⁴. In a broad sense, this film is a biological film, as many of the components including microscopic propagules of macrofouling organisms like bryozoans and ascidians are bioactive. Such bioactive film promotes cell attachment by modifying surface properties (for example, wettability, roughness, hydrophobicity, surface charge and surface free energy)^{4,18}, provides cells with more nutrients¹³ and protects them from predators¹⁹ or harsh environments^{20,21}.

The chemical composition of the conditioning film determines its physiochemical properties that affect cell attachment²⁰. Surface charge and hydrophobicity are two major factors that govern initial cell attachment. Carbohydrate polymers (for example, polysaccharides and lipids) are one of the most abundant constituents of the conditioning film that largely modify and determine surface charge and hydrophobicity^{4,20}. Amino acids, proteins and extracellular DNA

(eDNA) are mostly charged molecules that might impact cell adhesion by changing the surface charge^{4,21,22}. Notably, bacteria are often negatively charged and thus more readily attach to surfaces with positive to neutral charges²³. Similarly, cells with a hydrophobic envelope preferentially adhere to a hydrophobic surface and vice versa. A direct in situ real-time observation showed that early biofilm formation is substantially facilitated by calcium cation bridging or adsorption via polysaccharide-related compounds on cell surfaces¹⁵. Few studies elucidate how specific chemical components of the conditioning film affect cell attachment at the molecular level. Linking the expression of genes regulating cell attachment with the chemical composition of conditioning films would help explain chemical biology mechanisms underpinning bioactive processes at the early biofilm formation stage.

Biofilm formation

Once cells touch a surface, their physiology is regulated at the (post-) transcriptional level, including downregulating motility genes and upregulating genes that promote cell attachment. Transition of cells from planktonic to sessile lifestyles is usually regulated by signaling molecules through different signal transduction systems (Fig. 2). The well-known signaling molecules are cyclic dinucleotides or nucleotides that act as intracellular second messengers (for example, c-di-GMP, c-di-AMP, cGAMP and cAMP)^{6,24} and indole⁷ (Fig. 2a).

c-di-GMP is one of the best-studied signaling molecules for bacterial biofilm formation^{5,6}. This signaling molecule consists of two nucleotide moieties bound by 3' → 5' phosphodiester bridges and is metabolically regulated by diguanylate cyclases (DGCs) for c-di-GMP synthesis from guanosine-5'-triphosphate (GTP) and phosphodiesterases (PDEs) for c-di-GMP degradation to GMP or 5'-phosphoguanidyl- (3',5')-guanosine (5'-pGpG)⁵ (Fig. 2b). Generally, c-di-GMP binds to effector proteins, allosterically regulates their activities and initiates downstream signal transduction cascades, with a series of signal outputs, such as cell motility and extracellular polymeric substance (EPS) synthesis²⁴ (Fig. 2b). For example, c-di-GMP binding to the PilZ domain-containing protein YcgR brings about allosteric activation of

the protein and initiates downstream signal transduction to the target flagellar motor proteins (for example, MotA, FlhG and FlhM) responsible for motility in *Escherichia coli*²⁵. High levels of c-di-GMP trigger a local control module of DgcM and PdeR, which form a DGC–PDE pair that directly interacts with and activates the transcription factor MlrA, resulting in enhanced expression of the central curli regulator CsgD²⁶ (Fig. 2b). Interestingly, CsgD simultaneously induces the transcription of curli genes and *dgcC*, which stimulates the synthesis of curli and c-di-GMP, and the latter further activates EPS production by binding to the cellulose synthase BcsA–BcsB complex²⁷. Therefore, chemical biologists could make an effort to reduce cellular c-di-GMP levels or disrupt its signal transduction as potential therapies against biofouling. In contrast to c-di-GMP, the roles of c-di-AMP in regulating biofilm formation do not reach consensus in different bacterial species. For example, c-di-AMP binds to its receptor CabPA, which in turn interacts with VicR, a response regulator that increases the expression of *gtfB*, encoding a glucosyltransferase required by EPS production, thereby promoting biofilm formation of *Streptococcus mutans*²⁸ (Fig. 2c). However, accumulation of c-di-AMP in *Bacillus subtilis* inhibits the expression of biofilm genes (for example, *epsA*, *tapA* and *slrR*), probably by increasing the activity of the transcription factor SinR, which binds to their promoter regions, resulting in transcription repression²⁹ (Fig. 2c). The differentiation of c-di-AMP signaling outputs suggests that signaling molecule receptors determine functional outputs in different bacterial species. Thus, we infer that c-di-AMP regulation of biofilm formation depends on its receptors. However, further studies on the response of different bacteria to the same c-di-AMP–receptor complex are needed to validate this inference. c-di-AMP is synthesized by a membrane-bound diadenylate cyclase from two ATP molecules and degraded by PDE to the linear molecule 5'-phosphadenylyl adenosine. These two enzymes or ATP metabolism could be targeted for management.

Hybrid cGAMP synthesized by the dinucleotide cyclase DncV is a newly discovered signaling molecule that regulates virulence and chemotaxis in *Vibrio cholerae* O1 biovar El Tor³⁰. Horizontally transferred DncV in *E. coli* ECOR31 produces cGAMP and simultaneously inhibits cell adhesion and motility by post-transcriptional downregulation of the gene *csgD* and the flagellar regulon gene *flhD*, respectively³¹ (Fig. 2d). Although purified DncV mainly produces cGAMP, other cyclic dinucleotides (for example, c-di-GMP and c-di-AMP) and unknown molecule(s) are also produced. There is no direct evidence that cGAMP directly regulates biofilm formation. Studies applying synthetic cGAMP to a broad spectrum of bacteria might validate its roles in biofilm formation.

By contrast, together with the RNA polymerase-binding transcription factor DksA, guanosine pentaphosphate (pppGpp) and guanosine tetraphosphate (ppGpp) regulate production of extracellular adhesive proteins (for example, LapA and LapF) and enhance biofilm formation of *Pseudomonas putida* KT2440 by downregulating the expression of the regulator FleQ and upregulating the expression of the sigma factor RpoS³² (Fig. 2e). On the contrary, elevated levels of cAMP result in

a decrease in c-di-GMP levels and Psl (polysaccharide synthesis locus) exopolysaccharide content because binding of cAMP to the virulence factor regulator Vfr increases the expression of some PDEs (that is, DipA, RbdA and BifA)³³ (Fig. 2f). Evidence shows the cross-talk between cAMP and c-di-GMP through direct interaction of their effectors in *Shewanella putrefaciens* (Fig. 2g), synergistically supporting biofilm maintenance³⁴. Thus, induction of cellular cAMP accumulation could be employed in management of biofilms.

Interestingly, indole is another intercellular signaling molecule that regulates biofilm formation, and many bacteria produce large quantities of indole⁷. Thus, indole is a widespread signal for biofilm formation in the natural environment. The regulation of *E. coli* biofilm formation by indole is linked to repression of mobility, reduction of chemotaxis and induction of the QS autoinducer 1 sensor (SdiA)³⁵. Indole stimulates biofilm formation in environmental strains of *V. cholerae* by activating polysaccharide expression genes³⁶. An interesting feature of indole is that it can modulate biofilm formation of bacteria that do not produce it, such as *Pseudomonas aeruginosa*³⁷, *Listeria monocytogenes*³⁸ and *S. mutans*³⁹. Furthermore, indole with cyclo(Pro-Tyr), a signaling molecule produced by *Burkholderia unamae*, is involved in bidirectional cell-to-cell interspecific communication, which modulates mixed-species biofilm formation between indole-producing and indole-consuming species⁴⁰. These findings suggest that disrupting interspecific communication mechanisms involving indole could be used to combat biofouling.

In addition, a variety of small noncoding RNA (sRNA) species can bind to messenger RNA or transcriptional regulators and sequentially modulate cell adhesion by post-transcriptional regulation of flagellar genes or the genes responsible for the production of curli and exopolysaccharide⁴¹. The RNA chaperone Hfq and its associated sRNA species, ArcZ and SdsR, upregulate the RpoS protein, which activates the expression of CsgD and thus enhances biofilm formation of *Salmonella enterica* serovar Typhimurium⁴². Notably, some sRNA species can also regulate c-di-GMP levels by modulating the expression of DGCs and PDEs. For example, the noncoding sRNA species CsrB and CsrC antagonize the translational inhibitor CsrA, which binds to the promoters of genes responsible for synthesis of exopolysaccharide poly- β -1,6-*N*-acetyl-glucosamine (PGA) and c-di-GMP^{43,44}, and promote biofilm formation of *E. coli* K-12. These observations suggest that chemical synthesis of similar sRNA species that can bind to the receptors or synthases of the signaling molecules could be a potential tool for RNA management.

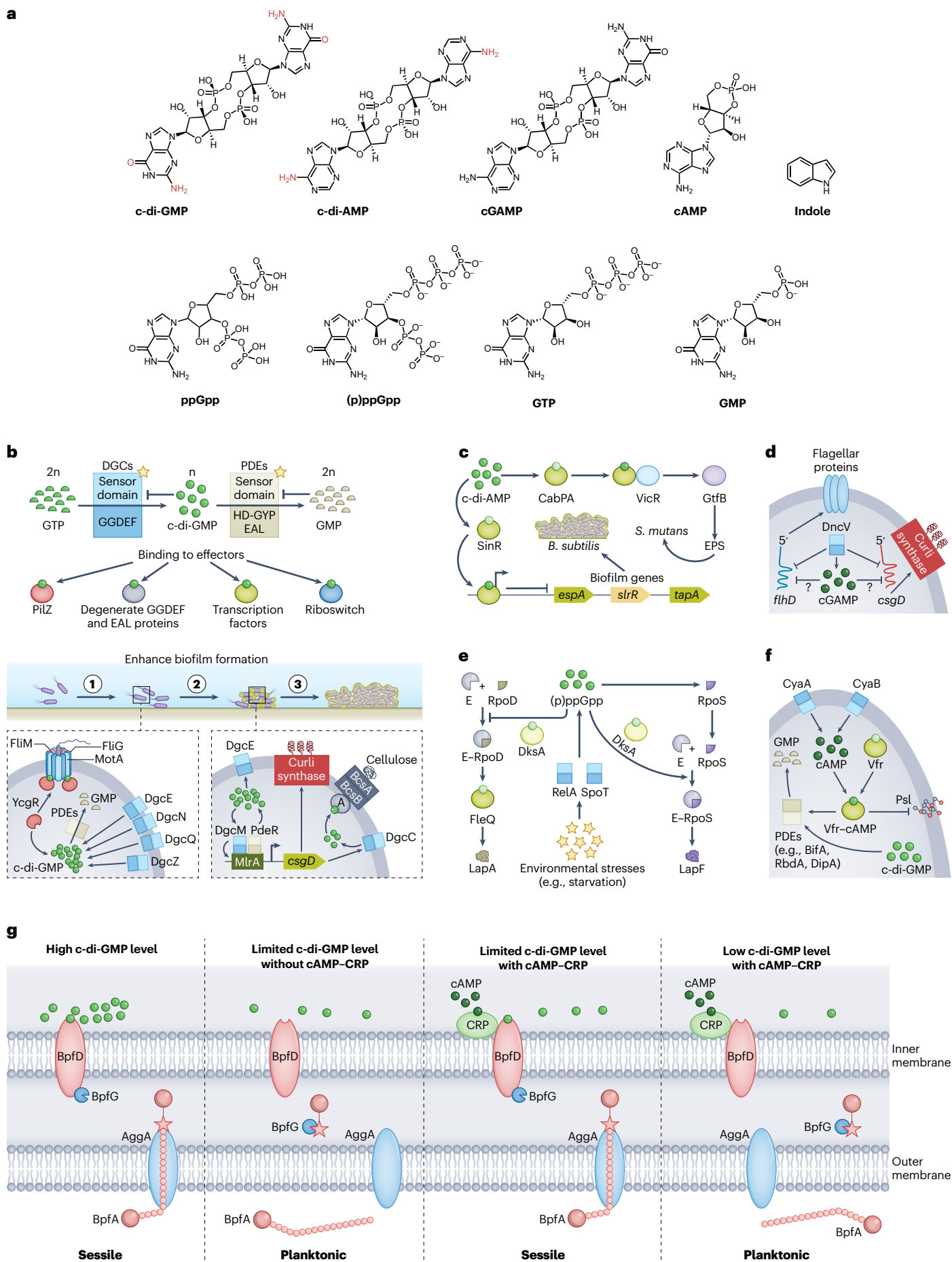
Although the literature on signaling molecules for biofilm formation is extensive, there are few studies on their exogenous interferents. On the basis of chemical biology⁴⁵, high-throughput screening of natural or synthetic chemicals capable of blocking signal transduction pathways would enable the development of novel antibiofilm agents.

Biofilm dispersion

In response to environmental stresses, such as limited nutrients or oxygen and other living conditions, mature biofilms start dispersion, with

Fig. 2 | Signaling molecules regulating biofilm formation. **a**, Structures of selected signaling molecules of cyclic dinucleotides. **b**, Enhancement of biofilm formation by c-di-GMP signaling. Cellular c-di-GMP levels are regulated by DGCs and PDEs. Transduction of c-di-GMP signals occurs through binding to its effectors, resulting in reduction of cell motility and an increase in EPS. Thus, higher c-di-GMP levels promote biofilm formation, whereas lower c-di-GMP levels reduce biofilm formation. GGDEF domain, a catalytic domain of DGCs that condenses two GTP molecules to a c-di-GMP molecule and is named after its conserved signature motif Gly-Gly-Asp-Glu-Phe. EAL domain, a catalytic domain of PDEs that hydrolyzes c-di-GMP into the linear molecule 5'-phosphoguananylyl-(3'-5')-guanosine (pGpG) and is named after its conserved signature motif Glu-Ala-Leu. HD-GYP domain, a catalytic domain of PDEs that cleaves c-di-GMP into two molecules of GM and belongs to a subclass of HD domain phosphohydrolases that contain an additional Gly-Tyr-Pro signature motif. HD domain is named after

the conserved doublet of predicted catalytic residues of phosphohydrolases. **c**, Inconsistence of c-di-AMP regulating biofilm formation in *S. mutans* and *B. subtilis*. **d**, Simultaneous inhibition of motility and cell adhesion of *E. coli* by cGAMP through post-transcriptional regulation of flagellar proteins and curli synthases, respectively. **e**, Enhancement of biofilm formation by (p)ppGpp through its effector, the RNA polymerase-binding transcription factor DksA. Environmental stresses (for example, starvation) activate the synthases (that is, RelA and SpoT) of (p)ppGpp, and its accumulation increases the extracellular adhesin LapF rather than LapA. **f**, Downregulation of biofilm formation by cAMP through its effector, the virulence factor regulator Vfr. The cAMP–Vfr complex can not only inhibit Psl exopolysaccharide production but also activate PDEs to degrade c-di-GMP. **g**, The BpfAGD system enables cross-talk between cAMP and c-di-GMP through direct interaction of their effectors, CRP and BpfD, respectively, resulting in biofilm maintenance. CRP, cAMP receptor proteins.



egress of single cells or disassembled cell aggregates (Fig. 3a). In brief, dispersion cues are sensed by the corresponding membrane-associated protein complexes, which in turn activate the production of signaling molecules that act on effectors, resulting in regulation of post-transcriptional expression of genes responsible for the group behaviors.

QS is a versatile mechanism of cell–cell communication, in which bacterial cells synchronously adjust the population density, resulting in biofilm formation when cell density is low (for example, on a pristine surface) or dispersion when cell density is high (for example, in mature biofilms). Thus, QS assembles planktonic cells to form an early biofilm, especially when very few cells are on the surface. Because of high cell densities in mature biofilms, QS, however, promotes biofilm dispersal. The mechanisms include reduction of EPS synthesis and an increase in bacterial motility through induction of low c-di-GMP levels, degradation of the biofilm matrix by upregulated EPS-degrading enzymes and production of biosurfactants that alter interfacial tension in biofilms¹⁶.

QS signaling molecules are usually called autoinducers, which mainly include *N*-acyl homoserine lactones (AHLs), furanosyl borate diesters (AI-2), *cis*-unsaturated fatty acids and autoinducing peptides (AIPs)^{46,47} (Fig. 3b). QS relies on the synthases and receptors of signaling molecules as well as the downstream signal transduction components. Bacteria usually use more than one QS system to regulate biofilm dispersal⁴⁷. For example, *P. aeruginosa* employs four QS systems using LasI and LasR, RhlI and RhlR, PqsABCDH and PqsR, AmbBCDE and an unknown receptor as the synthases and receptors of the autoinducers *N*-3-oxododecanoyl homoserine lactone (3OC₁₂-HSL), *N*-butanoyl-L-homoserine lactone (C₄-HSL), *Pseudomonas* quinolone signal (PQS) and 2-(2-hydroxyphenyl)-thiazole-4-carbaldehyde (IQS), respectively⁴⁸ (Fig. 3c). PQS (that is, 2-heptyl-3-hydroxy-4-quinolone) induces the formation of outer membrane vesicles (OMVs), which can actively degrade extracellular proteins, lipids and DNA, thus facilitating cell escape by coordinating the controlled degradation of biofilm matrix components⁴⁹.

Binding of RhlR with C₄-HSL enables activation of the *rhlA* promoter to initiate transcription of *rhlAB* genes to produce rhamnolipids, which are biosurfactants that promote biofilm dispersal by selectively increasing cell permeability⁵⁰ or by enhancing cell–cell interactions and preventing cells from attaching to surfaces⁵¹. After binding to 3OC₁₂-HSL, LasR upregulates expression of the tyrosine phosphatase TpbA, which represses *pelACDF* expression and decreases c-di-GMP levels, probably through inactivation of the DGC TpbB⁵². This directly leads to a reduction in Pel exopolysaccharide, required for biofilm formation. However, IQS is a potent biofilm dispersal signal that functions by integrating QS networks, including the central Las system and the downstream PQS and Rhl regulatory systems, with the phosphate stress-response mechanism⁵³. Similarly, *Vibrio* spp. have four QS systems with LuxM and LuxN, LuxS and LuxPQ, CqsA and CqsS, and Tdh and VqmA as the synthases and receptors of the autoinducers AI-1, AI-2, (S)-3-hydroxytridecan-4-one and 3,5-dimethylpyrazin-2-ol (DPO), respectively^{16,54} (Fig. 3d). At high cell densities, binding of AI-1, AI-2 and

(S)-3-hydroxytridecan-4-one to receptors converts them from kinases to phosphatases. The phosphatases dephosphorylate and inactivate LuxO and LuxU, resulting in termination of the expression of genes encoding four homologous regulatory sRNA species, Qrr1–Qrr4. The Qrr1 sRNA activates translation of the low-cell density master regulator AphA. AphA is a key transcriptional activator of biofilm formation and destabilizes messenger RNA of the high-cell density master regulator HapR, which inhibits translation of the transcriptional activator VpsT, required for biofilm formation⁵⁴. Similarly, DPO binds to the VqmA receptor and then activates transcription of the *vqmR* gene encoding the VqmR sRNA, which represses translation of VpsT, thus inducing biofilm dispersal⁵⁵. In *Staphylococcus aureus*, QS systems include the accessory gene regulator (Agr) system and the LuxS system (Fig. 3e). The Agr system uses AgrB as the synthase and the AgrD precursor peptide as the substrate to produce the AIP, which in turn signals through the AgrC receptor and the downstream transcription factor AgrA⁵⁶. Reactivation of the Agr system in mature *S. aureus* biofilms through AIP addition leads to AgrA phosphorylation, which triggers the expression of RNAIII, inducing the secretion of extracellular proteases that degrade proteins such as adhesins, contributing to dispersal⁵⁷. The LuxS system produces AI-2, which activates transcription of the intercellular adhesion (Ica) repressor (IcaR), which inhibits expression of the *ica* locus (that is, *icaA*, *icaD*, *icaB* and *icaC* in an operon) by binding to the promoter region of *icaA*, interrupting the secretion of extracellular polysaccharides⁵⁸. Further identification of potential QS receptors by chemical biologists would provide more biofilm dispersal strategies for management.

Some QS molecules are typical diffusible signal factors (DSFs), especially *cis*-unsaturated fatty acids, such as *cis*-2-decenoic acid, *cis*-11-methyl-2-dodecenoic acid (DSF) and *cis*-2-dodecenoic acid, which induce biofilm dispersal by reducing c-di-GMP levels or enhancing synthesis of enzymes that degrade the biofilm matrix¹⁷. In *Xanthomonas campestris*, the membrane-bound histidine kinase RpfC senses DSF produced by RpfF, resulting in dephosphorylation of RpfC and the release of RpfF, which in turn increases DSF production. The phosphate released from RpfC is transferred to RpfG, which activates its PDE activity and lowers c-di-GMP levels⁵⁹. The RpfC–RpfG two-component system represses expression of *xagABC*, encoding a glycosyltransferase system essential for synthesis of exopolysaccharides⁶⁰, and promotes the synthesis of *manA*-encoded endo-β-1,4-mannanase, which degrades the matrix⁶¹ (Fig. 3f). Thus, chemical synthesis of DSFs or their derivatives could be used as a proxy for dispersal signaling.

Some non-QS signals can induce biofilm dispersion by reducing cellular c-di-GMP concentrations and degrading the biofilm matrix. In response to glutamate, the chemotaxis-like methyl-accepting chemotaxis protein homolog BdlA activates DipA, a PDE, thus reducing c-di-GMP levels in *P. aeruginosa*⁶². Similarly, nitric oxide activates two membrane-bound proteins, NbdA and MucR, with PDE activity, resulting in low c-di-GMP levels⁶³ and reduced levels of Pel polysaccharide⁶⁴. These two non-QS signals can upregulate the expression of genes encoding the DNA-specific endonuclease EndA, which degrades eDNA in the matrix⁶⁵, and glycoside hydrolases (for example,

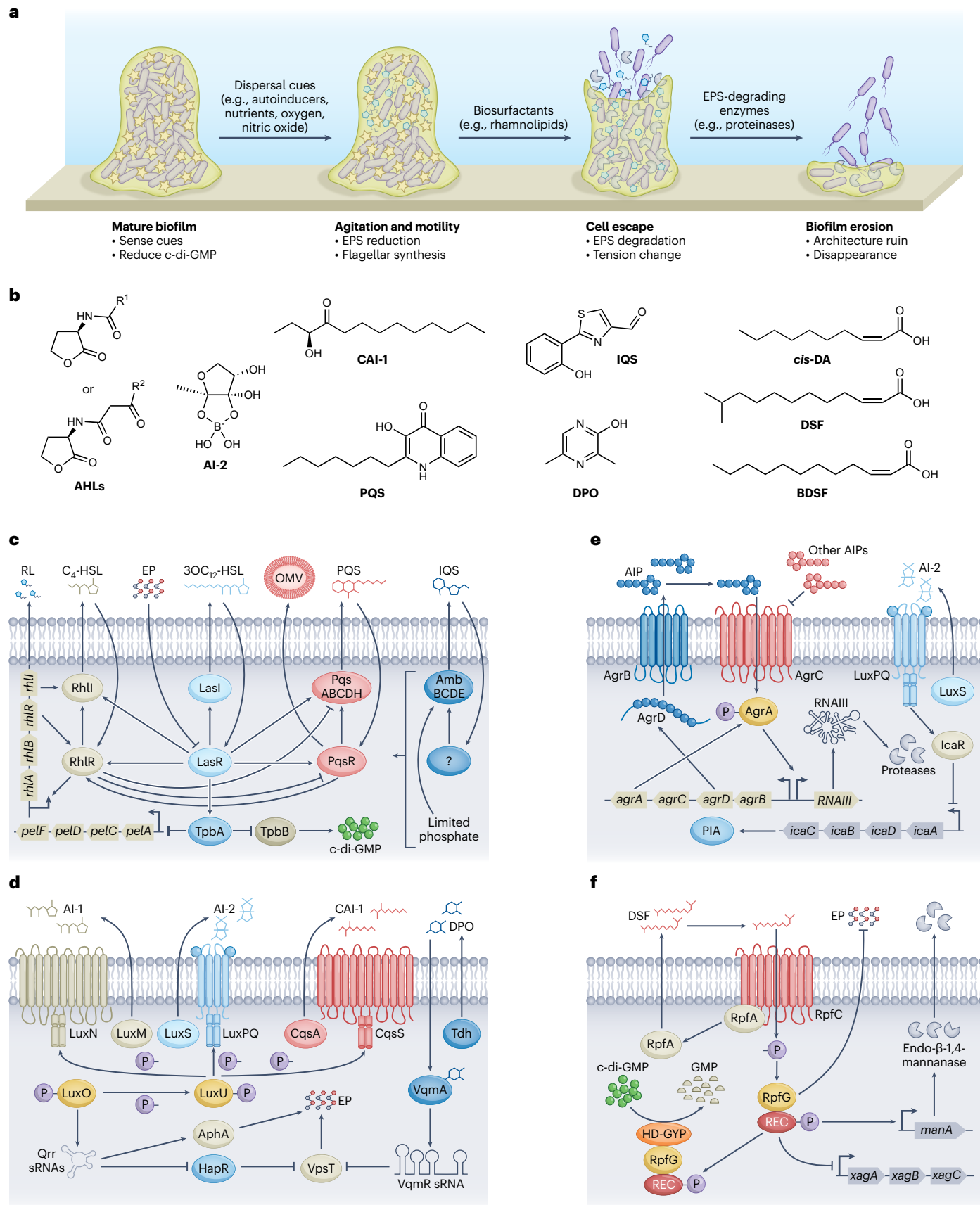
Fig. 3 | Signaling molecules inducing biofilm dispersion. **a**, Scheme of biofilm dispersal in response to environmental stresses. **b**, Structures of selected inducers of biofilm dispersion. Most of them are QS molecules because they are dispersal cues for a mature biofilm of high cell density. R¹ can be –C_nH_{2n+1} (*n* > 1) or other chemical groups, such as –C₃H₆OH for C₄-HSL, whereas R² groups were only found to be –C_nH_{2n+1}, such as –CH₃ for *N*-(3-oxobutanoyl)-L-homoserine lactone and –C₉H₁₉ for 3OC₁₂-HSL. BDSF, *cis*-2-dodecenoic acid; CAI-1, (S)-3-hydroxytridecan-4-one; *cis*-DA, *cis*-2-decenoic acid. **c**, QS systems regulating biofilm dispersal of *P. aeruginosa* by increasing biosurfactant (for example, rhamnolipid (RL)) production and/or reducing c-di-GMP synthesis. The central Las system interacts with downstream PQS and Rhl regulatory systems, with the IQS signaling system affecting the other three under the condition of limited

phosphate. **d**, QS systems regulating biofilm dispersal of *Vibrio* spp. On the one hand, accumulation of QS chemical cues leads to dephosphorylation and inactivation of LuxO and LuxU, resulting in reduction of exopolysaccharides (EP); on the other hand, binding of DPO to its receptor VqmA promotes the synthesis of the VqmA sRNA, which inhibits VpsT, required for exopolysaccharide production. **e**, Biofilm dispersal of *S. aureus* induced by the Agr and Lux QS systems by enhancing protease synthesis and reducing secretion of polysaccharide intercellular adhesives (PIA). **f**, DSFs promoting biofilm dispersal in *X. campestris*. Accumulation of DSFs promotes phosphorylation of RpfG, which leads to biodegradation of exopolysaccharides. Adapted from ref. 47, Springer Nature.

PslG and PelA) that degrade the exopolysaccharides Psl and Pel⁶⁶. A typical glycoside hydrolase is dispersin B (DspB), which is a soluble β -N-acetylglucosaminidase responsible for biofilm dispersion of bacteria, such as *Actinobacillus actinomycetemcomitans*⁶⁷. Given relatively

stable chemical properties and accessibility, such non-QS signals might allow for flexible management of biofouling in broader environments.

Although the perception of dispersion cues and the subsequent modulation of c-di-GMP concentrations have been well investigated,



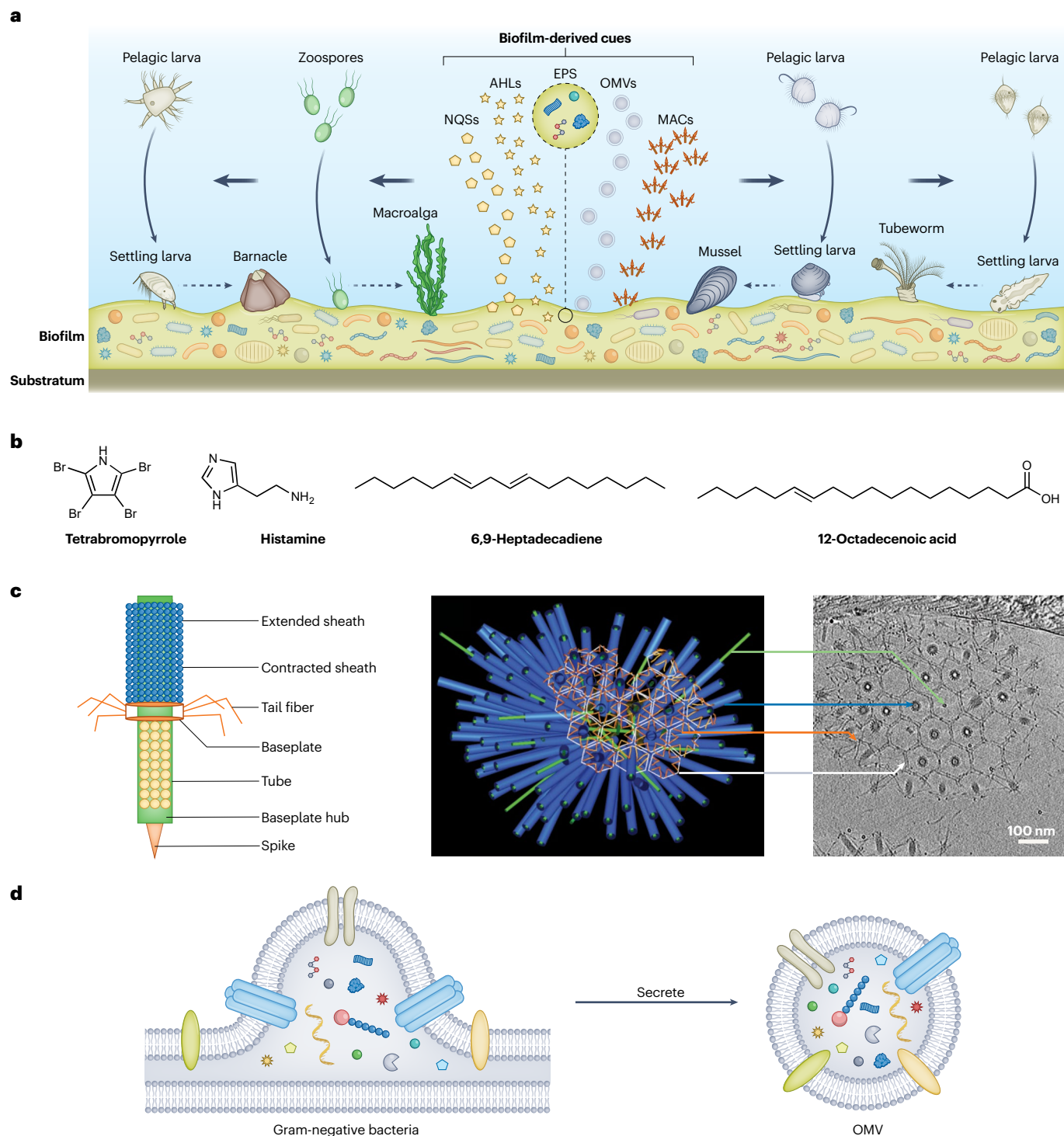


Fig. 4 | Biofilm induction of larval settlement. **a**, Overview of biofilm-induced macrofouling processes through a variety of water-soluble molecules, biomacromolecules and surface-bound EPS. Dashed lines represent metamorphosis from the larva to the adult. **b**, Structures of selected non-QS signals discovered in marine biofilms. **c**, Structure of phage tail-like MACs. MACs are high-molecular-weight bactericidal protein particles that resemble and are evolutionarily related to the tail structures of various bacteriophages. Adapted

with permission from ref. 77, AAAS. **d**, Secretion of OMVs as settlement cues. OMVs are small, spherically bilayered (100–300-nm) vesicles released into the extracellular milieu from the outer membrane of Gram-negative bacteria. They carry lipopolysaccharides, phospholipids, peptidoglycan, cell wall components, proteins (periplasmic, cytoplasmic and membrane bound), nucleic acids (DNA or RNA), ion metabolites and signaling molecules as cargo to a larva for settlement.

little is known about the events contributing to biofilm dispersion and the regulatory mechanisms of matrix degradation. It is unclear whether matrix degradation is the direct result of c-di-GMP concentrations or whether additional factors are required to activate the expression of genes encoding matrix-degrading enzymes¹⁷. Further contributions

from chemical biologists could focus on how cell behaviors such as motility and EPS production change in mature biofilms in response to specific dispersion signaling molecules. This direction of work would enable the discovery of more chemical derivatives for induction of biofilm dispersion.

Although extensive studies investigate the roles of the above-mentioned signaling molecules in biofilm development, the results obtained thus far are species specific and scattered. There are no universal rules that can be applied to predict the formation and dispersal patterns of natural multiple-species biofilms.

Biofilm-mediated marine macrofouling

Benthic invertebrates and macroalgae are ubiquitous sessile organisms. Typically, their planktonic larvae or zoospores enable distribution on a broader geographic scale and optimize settlement on surfaces for sessile juveniles and growth to adults. Preformed biofilms covering all aquatic interfaces determine larval and spore settlement of many but not all macrofouling species. Interactions between bacteria and macrofouler propagules are mediated by chemical cues, which can be water-soluble small molecules or insoluble biomacromolecules produced by biofilms or degradation products of biomacromolecules (Fig. 4a). These biofilm-derived cues have repulsive or attractive effects on macrofouling larvae¹², implying potential for effective anti-biofouling solutions. To this end, we focus on the chemical ecology of biofilm-derived cues to advance the understanding of macrofouling settlement behaviors.

Induction of larval settlement

Currently, biofilms elicit larval settlement through three major routes: (1) releasing water-soluble molecules, including QS signals and peptides generated by exogenous serine proteases, (2) secreting biomacromolecules, including proteins and hydrolytic enzymes, OMVs and phage tail bacteriocins and (3) relying on the surface-bound cues of EPS. Notably, such mechanisms usually co-occur in natural multispecies biofilms (Fig. 4a).

Water-soluble chemical cues that induce larval settlement are mainly QS molecules. QS signals from marine biofilms have been found predominantly in *Vibrio* spp. The first AHL molecule, *N*-(3-oxohexan-3-yl)-3-aminodihydro-2(3H)-furanone, was also discovered in the marine bacterium *Vibrio fischeri*⁶⁸. Further observations confirmed the presence of a QS system mediated by different AHLs (for example, *N*-butanoyl homoserine lactone and *N*-hexanoyl homoserine lactone) in marine biofilm communities⁶⁹. These observations suggest that QS signals widely occur in biofilms and marine microbial communities⁸. Thus, QS molecules have been subsequently studied as biofilm-derived cues to induce larval settlement due to cross-kingdom signaling. For example, AHLs produced by *Vibrio anguillarum*⁷⁰ or by two epiphytic bacteria, *Sulfitobacter* spp. and *Shewanella* spp.⁷¹, induce *Ulva* zoospore settlement and further germination and growth. Coincidentally, a plasticizer used in plastic water bottles, dibutyl phthalate, also induces *Ulva* zoospore settlement and development⁷², providing an example of potential management by chemical biology. A typical laboratory-based investigation has demonstrated that single-species biofilms of AHL-producing bacteria contain attractants for *Balanus improvisus* cyprid settlement⁷³. The ability to attract cyprids is abolished when AHL production is inactivated, whereas addition of synthetic AHLs enables biofilms to induce larval settlement. Similarly, larvae of the polychaete *Hydroides elegans* have been observed to shift from swimming to crawling behaviors in response to synthetic AHLs or biofilm-derived AHLs⁷⁴. However, a variety of studies from the 1980s on show that other barnacles settle in the absence of bacterial films and that some bacterial films have no effect. Moreover, non-QS soluble compounds, including peptides, are also identified as inducers of larval settlement (Fig. 4b). Tetrabromopyrrole is the first instance of a chemical cue isolated from a marine biofilm bacterium that induces the settlement of Caribbean coral larvae⁷⁵. Histamine released from algae has been identified as the settlement cue for the sea urchin *Holopneustes purpurascens*⁷⁶.

Some biomacromolecules secreted by bacteria in biofilms can also relay cues inductive to the settlement of larvae and spores. Identification of arrays of bacterial phage tail-like structures produced by a biofilm-forming bacterium, *Pseudoalteromonas luteoviolacea*,

has opened a window to understanding how marine biofilms trigger the larval settlement of the tubeworm *H. elegans*⁷⁷ (Fig. 4c). Further molecular-level studies have demonstrated that phage tail-like metamorphosis-associated contractile structures (MACs) trigger larval settlement and metamorphosis of *H. elegans* by inducing the sequential regulation of groups of genes responsible for tissue remodeling, innate immunity and mitogen-activated protein kinase (MAPK) signaling⁷⁸. The *H. elegans* story promises to have several more interesting chapters.

Moreover, bacterial OMVs, which have been defined as ‘sample packs’ to secrete parent cell wall contents (for example, peptidoglycan, lipopolysaccharide, periplasmic and membrane-bound proteins, enzymes, DNA and RNA)⁷⁹, are naturally universally produced by Gram-negative bacteria⁸⁰ (Fig. 4d). The bacterial contents in OMVs might be potential cues for settlement of marine invertebrates, considering their universal distribution in microbial communities. Inspired by this, some studies confirmed the presence of settlement-inductive fractions in OMVs and highlighted the importance of OMVs in eliciting invertebrate larval settlement⁸¹. Further investigation has also confirmed that lipopolysaccharide in OMVs produced by *Cellulophaga lytica* can induce larval settlement of *H. elegans*, suggesting that it might be a broadly employed cue due to its widespread prevalence⁸². Therefore, OMVs deliver a variety of chemical cues and act as a bridge between planktonic larvae and biofilms.

Apart from bacterial cells, EPS constitute the major fraction of the biofilm matrix. EPS are mainly polysaccharides, proteins, nucleic acids and lipids²¹, enabling the ecological success of biofilms as habitat colonizers⁸³. Macromolecules enable resource capture and retention of inorganic and organic particles, organic molecules and metal ions. Such highly complex chemical composition and function enable EPS to be a surface-bound cue to the settlement of invertebrate larvae (Fig. 4a). Early in 1985, evidence of lectins (that is, sugar-binding proteins and glycoproteins) in the settlement and metamorphosis of marine invertebrate larvae was implicated, in which lectins on the surface of the larvae can recognize and bind to a glycoconjugate of marine bacterial EPS⁸⁴. We previously showed that exopolysaccharides (that is, cellulose and colanic acid) of the bacterium *Persephonella marina* promote mussel larval settlement under the coordination of c-di-GMP^{85,86}. Further treatment of *P. marina* biofilm EPS with amylases, proteases and lipases decreased inducing activity, suggesting that inductive cues may be polysaccharides, proteins and lipids⁸⁷.

Some flagellar proteins derived from *P. marina* have been reported to trigger the settlement of *Mytilus coruscus*⁸⁸. Deletion of the gene *fliP*, responsible for flagellin synthesis, resulted in loss of flagellar proteins and reduced settlement, whereas biofilms of $\Delta fliP$ mutants had the same settlement-inducing activity when complemented with flagellar proteins. Importantly, the flagellar secretion system (F-T3SS) allows a flagellum to export its filament to the outside of the cell, which enables bacterial secretion of toxins and other proteins⁸⁹, suggesting the release of chemical cues from bacterial biofilms. However, how such proteinaceous signals mediate larval settlement remains elusive. Understanding of relationships between specific chemical cues and the expression of genes that regulate larval settlement will advance the explanation of molecular mechanisms underpinning the processes of biofilm-induced macrofouling.

Although such chemical cues from biofilms are considered to play important roles in larval settlement, the molecular mechanisms of how macrofouling organisms sense them are unclear. Metazoans are very complex organisms, and impacts can be from external receptors to any point in the transduction pathways that regulate metamorphosis.

Inhibition of larval settlement

To compete for space, precolonized biofilms could reject the settlement of subsequent colonizers (Fig. 5). This inhibitory effect is determined by the community structure of the biofilms and their active metabolism. An early study by Maki and his colleagues implied that films and

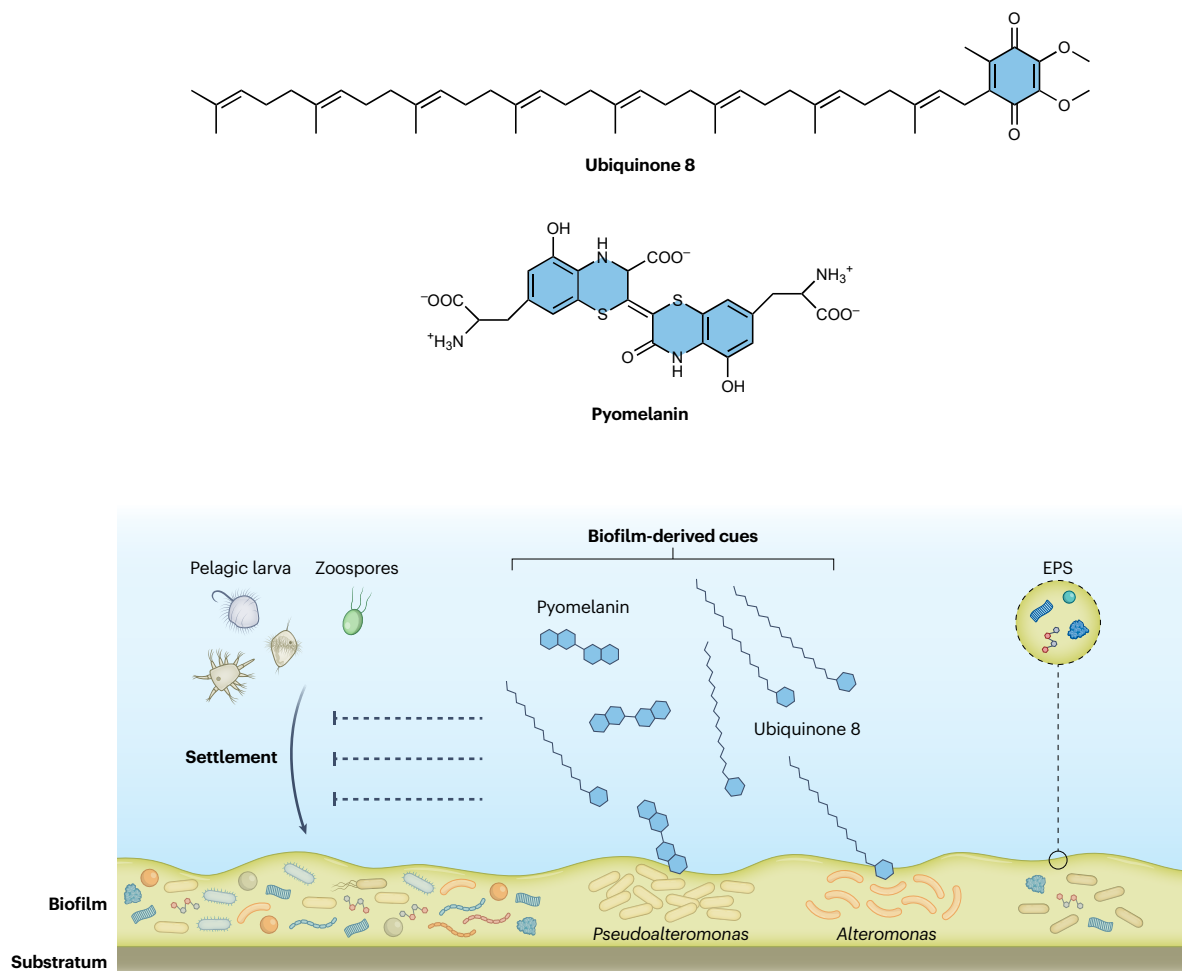


Fig. 5 | Biofilm inhibition of larval settlement. Preformed biofilms as pioneer colonizers also exhibit competition with latecomers for more niches. Biofilm members could produce metabolites that have inhibitory effects on larval

settlement. Simultaneously, a diversity of EPS enables the biofilms to avoid foreign competitors or opponents. Dashed lines represent metamorphosis from the larva to the adult.

exopolysaccharides of *Halomonas marina*⁹⁰ and *P. marina*^{91,92} may inhibit barnacle larval settlement. Such clues have greatly advanced further investigations of biofouling inhibition of microorganisms and their metabolites in the last decades.

Generally, most studies have focused on characterization of inhibitory effects of monospecies or multispecies biofilms and their EPS, whereas few have identified the specific inhibitors produced by biofilms. An early observation showed that biofilm exudates of *Macroccoccus* sp. AMGM1, *Bacillus* sp. AMGB1 and *Pseudoalteromonas* sp. AMGP1 result in inhibition of mussel (*Aulacomya maoriana* Iredale 1915) larval settlement⁹³. One of the early characterized inhibitors of barnacle cyprid settling is ubiquinone 8, produced by *Alteromonas* sp. KK10304 isolated from the marine sponge *Halichondria okadai*⁹⁴. Settlement inhibitors have been mainly identified from the genus *Pseudoalteromonas*, commonly found in various marine environments, especially *Pseudoalteromonas tunicata*⁹⁵. For example, our observation showed that pyomelanin secreted by the genus *Pseudoalteromonas* can inhibit larval settlement of the mussel *M. coruscus*⁹⁶, suggesting that pyomelanin might be a universal inhibitive cue released by marine biofilms.

However, recent advances on how invertebrate larvae sense chemical cues are limited. Currently, the settlement-inducing protein complex (SIPC), a macromolecular glycoprotein, has been discovered to act in cyprids as a pheromone to induce larval settlement and as an adhesive to explore the surface⁹⁷. An interesting observation indicated that domains at the N terminus of SIPC transduce settlement-preference cues that mask the settlement-avoidance cues

transduced by domains at the C terminus in *Amphibalanus amphitrite*¹¹. Moreover, the signal transduction pathway for the SIPC effects permanent attachment through secretion of cyprid cement, whereas removal of the N-terminal 600 amino acids from the SIPC results in transduction of only settlement-avoidance cues to the cyprids. The settlement-inducing effect of the SIPC can be mimicked by designing a variety of small 9–3-residue peptides ending in arginine or lysine⁹⁸. Despite this, molecular mechanisms of how larvae recognize biofilm-derived cues and subsequently translate into settlement behaviors are less clear. Again, metazoans are very complex, and most biofouling organisms metamorphose from planktonic propagules to sessile juveniles. Metamorphosis is a complex and highly integrated process with many steps.

To unveil the mechanisms underpinning biofilm-mediated macrofouling processes, contributions from chemical biologists should be made to (1) identify chemical cues and their corresponding effector proteins universal to different species, (2) envision chemical–protein reaction models to uncover the signal transduction pathways, and (3) demonstrate pathways by applying the principles and tools of chemistry. Furthermore, these efforts could produce a paradigm that works on a broader spectrum of chemical cues and even on the signal transduction pathways between microorganisms and hosts⁹⁹.

Promising anti-marine macrofouling approaches

Because biofilms have a fundamental role in marine biofouling, we argue that anti-biofouling strategies should turn to the chemical

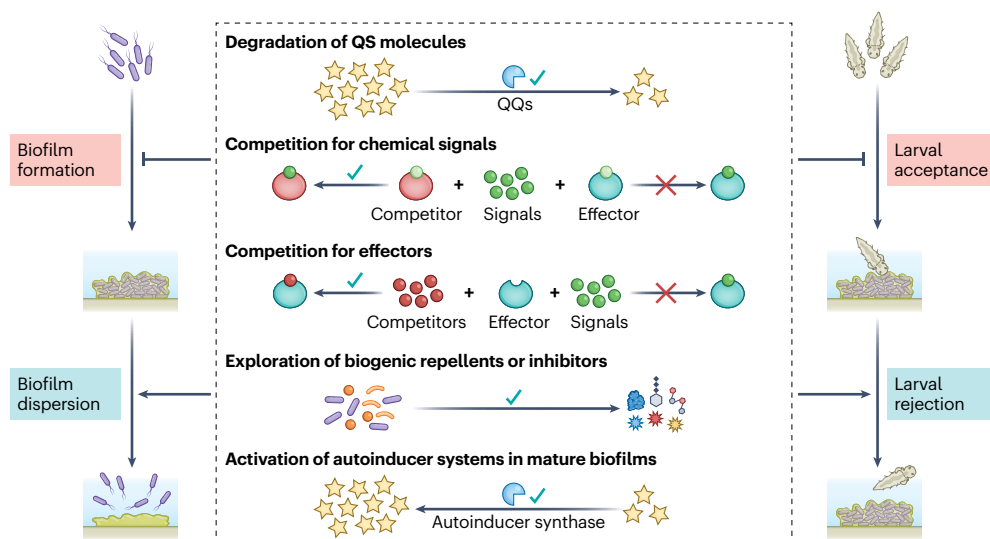


Fig. 6 | Potential chemical biology strategies for anti-biofouling. Biofouling problems stem from biofilm formation, and thus anti-biofouling strategies should focus on biofilm control. Any biochemical processes that inhibit biofilm formation, enhance biofilm dispersal, reduce larval settlement or enhance larval

rejection could be explored as a potential anti-biofouling solution. Lines with a green tick mark represent enhancement or promotion of the corresponding processes.

biology of biofilms. Potential solutions toward sustainability would underpin the processes that interfere with biofouling. Therefore, studies on the chemical ecology of signaling molecules responsible for biofilm formation and dispersal would allow the development of innovative anti-biofouling approaches (Fig. 6).

Quorum quenching disrupts QS signaling and is widely proposed as an anti-biofouling strategy¹⁰⁰. High-throughput screening of quorum quenching molecules (for example, quorum quenching enzymes and QS inhibitors) that degrade QS signaling molecules or impede cellular signaling cascades might be effective. As discussed above, high c-di-GMP levels are conducive to EPS production and biofilm formation, while low c-di-GMP levels result in planktonic growth and/or biofilm dispersion. c-di-GMP concentrations could be used as a proxy for inducing biofilm dispersal. Unfortunately, activating PDEs or inactivating DGCs does not necessarily work because there are always multiple PDEs or DGCs in bacteria. However, consumption or obstruction of the c-di-GMP signal through chemical synthesis of invalid effectors can be used as a strategy for competing with the c-di-GMP effectors. Based on the chemical structures of the effectors, efforts from chemical biology could be focused on designing and identifying false effectors or receptors that are much more competitive for the signaling molecules. Largely inspired by inhibitory effects of microorganisms, biodegradable or biogenic compounds (for example, small molecules, peptides and proteins) have been explored as natural antibiofilm agents and repellents¹⁰¹. For example, inhibitory effects on biofilm formation of *P. aeruginosa*, *E. coli*, *Acinetobacter baumannii* and *Klebsiella pneumoniae* have been achieved with the potent antibiofilm peptide 1018 by blocking (p)ppGpp signaling¹⁰². These biologically based repellents would open new avenues for the development of nonbiocidal therapies for biofouling issues.

A central question is: can a broad-spectrum solution be found that does not rely on biocides? The natural environment is diverse and dynamic, and organisms select a suitable surface to settle and grow on. There are a finite number of ways that biofouling organisms function. Several patents describe small molecules that interfere with macrofouler metamorphosis. The interesting aspect in this case is that, although the various macrofoulers appear different and metamorphose differently as a result of the target molecules, somehow the metamorphic transduction cascade is interrupted. The key here is using additives that are protected in a fouling-management coating

and degrade quickly upon release. We expect that molecules will be discovered that impact other groups of biofouling organisms. Considering a broad role of QS, indole and c-di-GMP signaling in biofilm formation, we argue that more attention should be paid to methods of interfering with biofilm-forming processes. Although targeting enzymes that degrade specific signaling molecules has potential, the instability of enzyme activity in changing environments is a challenge. By contrast, design and synthesis of green chemicals (for example, peptides²⁵) that react with signaling molecules to impede cellular signaling cascades could be more effective for sustainable mitigation strategies¹⁰¹. Notably, biofilm dispersal does not mean full dispersal; only a fraction of the biofilm-associated bacteria becomes dispersed. Thus, further studies are needed to test whether or not the remaining surface-associated bacteria are sufficient to promote macrofouling before a specific dispersal strategy is proved to be successful in combatting marine fouling. If yes, chemical biologists could refocus on application of small molecules produced through synthetic chemistry to disrupt signaling from the remaining biofilms or to weaken the signal intensity below effective levels. Again, indole derivatives (for example, 6-chloroindole¹⁰³, *N*-(1-*H*-2-phenyl-indole-3-ylmethyl) benzamide¹⁰⁴ and indole, 3-formylindole and cyclo(Pro-Leu)¹⁰⁵) produced naturally as bacterial secondary metabolites or synthesized chemically are promising alternatives due to their biocidal and antibiofilm activity, low toxicity and low bioaccumulation.

Alternatively, other anti-fouling approaches could be incorporated. For instance, engineering the characteristics of surfaces to modulate organisms for settlement and bacterial biofilms may achieve novel solutions to interfere with surface sensing and/or deter community formation¹⁰⁶. Combined with such implications, a hybrid solution could be available in practice, such as a novel hydraulic pressure-responsive membrane designed for in situ self-cleaning of a broad range of foulants through reactive oxygen species production and dielectrophoretic repulsion¹⁰⁷.

Conclusions and future perspectives

Over the past few decades, studies have been focused on biofilm development of a few model species, and signaling molecules involved in microfouling are relatively clear. However, the development of multi-species biofilms, where interspecies and cross-kingdom communication prevails in complex natural environments, needs to be understood.

Although biofilms are commonly known to recruit or repel macroorganisms through chemical cues, the molecular mechanisms underlying biofilm–larva interactions are unclear. Macrofouling settlement is so essential to survival that there are multiple redundant triggers and a finite number of pathways and processes. In nature, biofilms are composed of multispecies consortia that include bacteria, fungal ciliates, diatoms, algae and invertebrates. We lack knowledge about how microorganisms interact and communicate with each other in natural biofilms and how they respond to environmental changes. Research on interactions between natural biofilms and macrofoulers is needed. Attempts to use natural biofilms as a model microbiome might help elucidate the mechanisms of how multispecies develop into a biofilm in clearly defined natural environments as the basis.

Although the literature on bacterial biofilm formation and dispersal is extensive, the majority of the mechanistic data are from laboratory-based models and clinically relevant strains. It is clear that many of the insights cannot be fully translated to marine biofilm-forming bacteria, contextualizing relevant regulation within natural multispecies biofilms. The complex regulatory networks of nucleotide synthesis make this connection difficult to generalize. Complications include: (1) contradictory data in the literature about the role of second messengers in different species, and (2) complicated regulatory motifs controlling cytoplasmic concentrations of c-di-GMP, including numerous PDEs, DGCs and the effects of other nucleotides.

While some aspects of biofilm regulation, such as secretion of QS compounds, are conserved, more fundamental research on the details of the signaling pathways of biofilm formation is needed to establish translation to marine organisms. Considering the high biodiversity and dynamic nature of marine biofilms^{3,9}, research priorities might be given to representative organisms, such as *Pseudoalteromonas* spp. and *Vibrio* spp., focusing on identifying the chemical cues liberated and elucidating their specific roles in regulating the macrofouling behaviors of model macrofoulers. Chemical synthesis of analogs of signaling molecules or chemical cues would allow the identification of signaling receptors, effectors and responsive species in marine environments. Modern technologies of multiomics (for example, proteomics and metabolomics) and microbiomes will help elucidate interactions between biofilms and macroorganisms.

Although a variety of surface-associated factors affect bacterial signaling and surface colonization¹⁰⁸, chemical cues merit more attention in biofilm-mediated biofouling. This is essential to understanding biofilm community dynamics and may offer important clues to anti-biofouling development. We also argue for the need to integrate larval ecology, microbial ecology, analytical chemistry and surface physics to measure the natural distribution and abundance of larval settlement cues and elucidate their ecological importance and niche specificity¹⁰¹. Thus, further research for environmentally benign anti-biofouling strategies would benefit from a multidisciplinary approach of chemical biology, molecular biology, physical chemistry and materials science to advance the fundamentals of biofouling and prevention.

References

- Cámara, M. et al. Economic significance of biofilms: a multidisciplinary and cross-sectoral challenge. *NPJ Biofilms Microbiomes* **8**, 42 (2022).
- Hofer, U. The cost of biofilms. *Nat. Rev. Microbiol.* **20**, 445 (2022).
- Antunes, J., Leão, P. & Vasconcelos, V. Marine biofilms: diversity of communities and of chemical cues. *Environ. Microbiol. Rep.* **11**, 287–305 (2019).
- Garg, A., Jain, A. & Bhosle, N. B. Chemical characterization of a marine conditioning film. *Int. Biodeterior. Biodegrad.* **63**, 7–11 (2009).
- Liu, X., Cao, B., Yang, L. & Gu, J.-D. Biofilm control by interfering with c-di-GMP metabolism and signaling. *Biotechnol. Adv.* **56**, 107915 (2022).
- Jenal, U., Reinders, A. & Lori, C. Cyclic di-GMP: second messenger extraordinaire. *Nat. Rev. Microbiol.* **15**, 271–284 (2017).
- Lee, J.-H. & Lee, J. Indole as an intercellular signal in microbial communities. *FEMS Microbiol. Rev.* **34**, 426–444 (2010).
- Hmelo, L. R. Quorum sensing in marine microbial environments. *Ann. Rev. Mar. Sci.* **9**, 257–281 (2017).
- Qian, P.-Y., Cheng, A., Wang, R. & Zhang, R. Marine biofilms: diversity, interactions and biofouling. *Nat. Rev. Microbiol.* **20**, 671–684 (2022).
- Dobretsov, S. & Rittschof, D. Love at first taste: induction of larval settlement by marine microbes. *Int. J. Mol. Sci.* **21**, 731 (2020).
- Kotsiri, M. et al. Should I stay or should I go? The settlement-inducing protein complex guides barnacle settlement decisions. *J. Exp. Biol.* **221**, jeb185348 (2018).
- Rittschof, D. et al. Cues and context: larval responses to physical and chemical cues. *Biofouling* **12**, 31–44 (1998).
- Siboni, N., Lidor, M., Kramarsky-Winter, E. & Kushmaro, A. Conditioning film and initial biofilm formation on ceramics tiles in the marine environment. *FEMS Microbiol. Lett.* **274**, 24–29 (2007).
- Rummel, C. D. et al. Conditioning film and early biofilm succession on plastic surfaces. *Environ. Sci. Technol.* **55**, 11006–11018 (2021).
- Chen, M.-Y. et al. Towards real-time observation of conditioning film and early biofilm formation under laminar flow conditions using a quartz crystal microbalance. *Biochem. Eng. J.* **53**, 121–130 (2010).
- Solano, C., Echeverez, M. & Lasa, I. Biofilm dispersion and quorum sensing. *Curr. Opin. Microbiol.* **18**, 96–104 (2014).
- Rumbaugh, K. P. & Sauer, K. Biofilm dispersion. *Nat. Rev. Microbiol.* **18**, 571–586 (2020).
- This review discusses conditions that lead to dispersion and the mechanisms by which native and environmental cues contribute to biofilm dispersion.**
- Bakker, D. P. et al. Multiple linear regression analysis of bacterial deposition to polyurethane coatings after conditioning film formation in the marine environment. *Microbiology* **150**, 1779–1784 (2004).
- Li, S., Liu, S. Y., Chan, S. Y. & Chua, S. L. Biofilm matrix cloaks bacterial quorum sensing chemoattractants from predator detection. *ISME J.* **16**, 1388–1396 (2022).
- This work describes an anti-detection adaptation, termed ‘biofilm cloak’, in which bacterial biofilms produce exopolysaccharides to reduce the leaching of autoinducers into the milieu, thereby concealing their trails to avoid detection by the predator. This adaptation indicates that restriction of bacterial signaling to the confinement of biofilms is a novel approach for predator evasion.**
- Jain, A. & Bhosle, N. B. Biochemical composition of the marine conditioning film: implications for bacterial adhesion. *Biofouling* **25**, 13–19 (2009).
- Flemming, H.-C. & Wingender, J. The biofilm matrix. *Nat. Rev. Microbiol.* **8**, 623–633 (2010).
- Lorite, G. S. et al. The role of conditioning film formation and surface chemical changes on *Xylella fastidiosa* adhesion and biofilm evolution. *J. Colloid Interface Sci.* **359**, 289–295 (2011).
- Berne, C., Ellison, C. K., Ducret, A. & Brun, Y. V. Bacterial adhesion at the single-cell level. *Nat. Rev. Microbiol.* **16**, 616–627 (2018).
- Kalia, D. et al. Nucleotide, c-di-GMP, c-di-AMP, cGMP, cAMP, (p) ppGpp signaling in bacteria and implications in pathogenesis. *Chem. Soc. Rev.* **42**, 305–341 (2013).
- Hou, Y. J. et al. Structural insights into the mechanism of c-di-GMP-bound YcgR regulating flagellar motility in *Escherichia coli*. *J. Biol. Chem.* **295**, 808–821 (2020).

26. Lindenberg, S., Klauck, G., Pesavento, C., Klauck, E. & Hengge, R. The EAL domain protein YciR acts as a trigger enzyme in a c-di-GMP signalling cascade in *E. coli* biofilm control. *EMBO J.* **32**, 2001–2014 (2013).
27. Morgan, J. L. W., McNamara, J. T. & Zimmer, J. Mechanism of activation of bacterial cellulose synthase by cyclic di-GMP. *Nat. Struct. Mol. Biol.* **21**, 489–496 (2014).
This work unveils crystal structures of the c-di-GMP-activated BcsA–BcsB complex by which c-di-GMP activates cellulose synthase, shedding light on mechanisms of c-di-GMP allosterically modulating enzymatic functions.
28. Peng, X., Zhang, Y., Bai, G., Zhou, X. & Wu, H. Cyclic di-AMP mediates biofilm formation. *Mol. Microbiol.* **99**, 945–959 (2016).
This study reveals a new role of c-di-AMP in mediating biofilm formation through a CabPA–VicR–GtfB signaling network in *S. mutans*.
29. Gundlach, J., Rath, H., Herzberg, C., Mäder, U. & Stülke, J. Second messenger signaling in *Bacillus subtilis*: accumulation of cyclic di-AMP inhibits biofilm formation. *Front. Microbiol.* **7**, 804 (2016).
30. Davies, B. W., Bogard, R. W., Young, T. S. & Mekalanos, J. J. Coordinated regulation of accessory genetic elements produces cyclic di-nucleotides for *V. cholerae* virulence. *Cell* **149**, 358–370 (2012).
31. Li, F. et al. DncV synthesizes cyclic GMP–AMP and regulates biofilm formation and motility in *Escherichia coli* ECOR31. *mBio* **10**, e02492-18 (2019).
32. Liu, H., Xiao, Y., Nie, H., Huang, Q. & Chen, W. Influence of (p) ppGpp on biofilm regulation in *Pseudomonas putida* KT2440. *Microbiol. Res.* **204**, 1–8 (2017).
33. Almlblad, H. et al. High levels of cAMP inhibit *Pseudomonas aeruginosa* biofilm formation through reduction of the c-di-GMP content. *Microbiology* **165**, 324–333 (2019).
34. Liu, C. et al. cAMP and c-di-GMP synergistically support biofilm maintenance through the direct interaction of their effectors. *Nat. Commun.* **13**, 1493 (2022).
This study provides distinctive evidence of cross-talk between cAMP and c-di-GMP pathways through direct interaction with their effectors, which biofilms maintain dynamically, to a certain extent.
35. Lee, J., Maeda, T., Hong, S. H. & Wood, T. K. Reconfiguring the quorum-sensing regulator SdiA of *Escherichia coli* to control biofilm formation via indole and *N*-acetylhomoserine lactones. *Appl. Environ. Microbiol.* **75**, 1703–1716 (2009).
36. Mueller, R. S., Beyhan, S., Saini, S. G., Yildiz, F. H. & Bartlett, D. H. Indole acts as an extracellular cue regulating gene expression in *Vibrio cholerae*. *J. Bacteriol.* **191**, 3504–3516 (2009).
37. Lee, J., Jayaraman, A. & Wood, T. K. Indole is an inter-species biofilm signal mediated by SdiA. *BMC Microbiol.* **7**, 42 (2007).
38. Rattanaphan, P., Mittraparp-Arthorn, P., Srinoun, K., Vuddhakul, V. & Tansila, N. Indole signaling decreases biofilm formation and related virulence of *Listeria monocytogenes*. *FEMS Microbiol. Lett.* **367**, fnaa116 (2020).
39. Inaba, T., Obana, N., Habe, H. & Nomura, N. Biofilm formation by *Streptococcus mutans* is enhanced by indole via the quorum sensing pathway. *Microbes Environ.* **35**, ME19164 (2020).
40. Hashidoko, Y. & Kim, D. Bidirectional cell–cell communication via indole and cyclo(Pro-Tyr) modulates interspecies biofilm formation. *Appl. Environ. Microbiol.* **87**, e01277-21 (2021).
41. Mika, F. & Hengge, R. Small RNAs in the control of RpoS, CsgD, and biofilm architecture of *Escherichia coli*. *RNA Biol.* **11**, 494–507 (2014).
42. Monteiro, C. et al. Hfq and Hfq-dependent small RNAs are major contributors to multicellular development in *Salmonella enterica* serovar Typhimurium. *RNA Biol.* **9**, 489–502 (2012).
43. Jackson, D. W. et al. Biofilm formation and dispersal under the influence of the global regulator CsrA of *Escherichia coli*. *J. Bacteriol.* **184**, 290–301 (2002).
44. Carzaniga, T. et al. Different *csrA* expression levels in C versus K-12 *E. coli* strains affect biofilm formation and impact the regulatory mechanism presided by the CsrB and CsrC small RNAs. *Microorganisms* **9**, 1010 (2021).
45. Yang, L. & Givskov, M. Chemical biology strategies for biofilm control. *Microbiol. Spectr.* **3**, MB-0019-2015 (2015).
46. Defoirdt, T. Amino acid-derived quorum sensing molecules controlling the virulence of vibrios (and beyond). *PLoS Pathog.* **15**, e1007815 (2019).
47. Mukherjee, S. & Bassler, B. L. Bacterial quorum sensing in complex and dynamically changing environments. *Nat. Rev. Microbiol.* **17**, 371–382 (2019).
48. Wellington, S., Greenberg, E. P., Winans, S. C., Blackwell, H. & Venturi, V. Quorum sensing signal selectivity and the potential for interspecies cross talk. *mBio* **10**, e00146-19 (2019).
This study investigated the selectivity of QS receptors and indicated ubiquitous mechanisms of substantial cross-talk between QS systems, which provides a basis for predictions about interspecies interactions in complex microbial communities.
49. Cooke, A. C. et al. *Pseudomonas* quinolone signal-induced outer membrane vesicles enhance biofilm dispersion in *Pseudomonas aeruginosa*. *mSphere* **5**, e01109-20 (2020).
50. Bhattacharjee, A., Nusca, T. D. & Hochbaum, A. I. Rhamnolipids mediate an interspecies biofilm dispersal signaling pathway. *ACS Chem. Biol.* **11**, 3068–3076 (2016).
51. Davey, M. E., Caiazza, N. C. & O'Toole, G. A. Rhamnolipid surfactant production affects biofilm architecture in *Pseudomonas aeruginosa* PAO1. *J. Bacteriol.* **185**, 1027–1036 (2003).
52. Ueda, A. & Wood, T. K. Connecting quorum sensing, c-di-GMP, Pel polysaccharide, and biofilm formation in *Pseudomonas aeruginosa* through tyrosine phosphatase TpbA (PA3885). *PLoS Pathog.* **5**, e1000483 (2009).
53. Lee, J. et al. A cell–cell communication signal integrates quorum sensing and stress response. *Nat. Chem. Biol.* **9**, 339–343 (2013).
54. Herzog, R., Peschek, N., Fröhlich, K. S., Schumacher, K. & Papenfort, K. Three autoinducer molecules act in concert to control virulence gene expression in *Vibrio cholerae*. *Nucleic Acids Res.* **47**, 3171–3183 (2019).
55. Papenfort, K. et al. A *Vibrio cholerae* autoinducer–receptor pair that controls biofilm formation. *Nat. Chem. Biol.* **13**, 551–557 (2017).
This work identified a new QS autoinducer, DPO, and characterized its roles in regulation of biofilm formation and toxin production, together with its receptor VqmA.
56. Luong, T., Sau, S., Gomez, M., Lee, J. C. & Lee, C. Y. Regulation of *Staphylococcus aureus* capsular polysaccharide expression by *agr* and *sarA*. *Infect. Immun.* **70**, 444–450 (2002).
57. Boles, B. R. & Horswill, A. R. *agr*-mediated dispersal of *Staphylococcus aureus* biofilms. *PLoS Pathog.* **4**, e1000052 (2008).
58. Yu, D., Zhao, L., Xue, T. & Sun, B. *Staphylococcus aureus* autoinducer-2 quorum sensing decreases biofilm formation in an icaR-dependent manner. *BMC Microbiol.* **12**, 288 (2012).
59. Ryan, R. P. et al. Cell–cell signaling in *Xanthomonas campestris* involves an HD-GYP domain protein that functions in cyclic di-GMP turnover. *Proc. Natl Acad. Sci. USA* **103**, 6712–6717 (2006).
60. Tao, F., Swarup, S. & Zhang, L.-H. Quorum sensing modulation of a putative glycosyltransferase gene cluster essential for *Xanthomonas campestris* biofilm formation. *Environ. Microbiol.* **12**, 3159–3170 (2010).
61. Dow, J. M. et al. Biofilm dispersal in *Xanthomonas campestris* is controlled by cell–cell signaling and is required for full virulence to plants. *Proc. Natl Acad. Sci. USA* **100**, 10995–11000 (2003).

62. Basu Roy, A. & Sauer, K. Diguanylate cyclase NicD-based signalling mechanism of nutrient-induced dispersion by *Pseudomonas aeruginosa*. *Mol. Microbiol.* **94**, 771–793 (2014).
This work provides a new DGC-based signaling mechanism that induces biofilm dispersion for biofilm control with glutamate as an inducer.
63. Li, Y., Heine, S., Entian, M., Sauer, K. & Frankenberg-Dinkel, N. NO-induced biofilm dispersion in *Pseudomonas aeruginosa* is mediated by an MHYT domain-coupled phosphodiesterase. *J. Bacteriol.* **195**, 3531–3542 (2013).
64. Hickman, J. W. & Harwood, C. S. Identification of FleQ from *Pseudomonas aeruginosa* as a c-di-GMP-responsive transcription factor. *Mol. Microbiol.* **69**, 376–389 (2008).
65. Cherny, K. E. & Sauer, K. *Pseudomonas aeruginosa* requires the DNA-specific endonuclease EndA to degrade extracellular genomic DNA to disperse from the biofilm. *J. Bacteriol.* **201**, e00059-19 (2019).
66. Cherny, K. E. & Sauer, K. Untethering and degradation of the polysaccharide matrix are essential steps in the dispersion response of *Pseudomonas aeruginosa* biofilms. *J. Bacteriol.* **202**, e00575-19 (2020).
67. Ramasubbu, N., Thomas, L. M., Ragunath, C. & Kaplan, J. B. Structural analysis of dispersin B, a biofilm-releasing glycoside hydrolase from the periodontopathogen *Actinobacillus actinomycetemcomitans*. *J. Mol. Biol.* **349**, 475–486 (2005).
68. Eberhard, A. et al. Structural identification of autoinducer of *Photobacterium fischeri* luciferase. *Biochemistry* **20**, 2444–2449 (1981).
69. Cuadrado-Silva, C. T., Castellanos, L., Arévalo-Ferro, C. & Osorno, O. E. Detection of quorum sensing systems of bacteria isolated from fouled marine organisms. *Biochem. Syst. Ecol.* **46**, 101–107 (2013).
70. Tait, K. et al. Disruption of quorum sensing in seawater abolishes attraction of zoospores of the green alga *Ulva* to bacterial biofilms. *Environ. Microbiol.* **7**, 229–240 (2005).
71. Twigg, M. S., Tait, K., Williams, P., Atkinson, S. & Cámara, M. Interference with the germination and growth of *Ulva* zoospores by quorum-sensing molecules from *Ulva*-associated epiphytic bacteria. *Environ. Microbiol.* **16**, 445–453 (2014).
72. Agusman et al. Conspecific cues that induce spore settlement in the biofouling and green tide-forming alga *Ulva tepida* provide a potential aggregation mechanism. *Int. Biodeterior. Biodegrad.* **145**, 104807 (2019).
73. Tait, K. & Havenhand, J. Investigating a possible role for the bacterial signal molecules *N*-acylhomoserine lactones in *Balanus improvisus* cyprid settlement. *Mol. Ecol.* **22**, 2588–2602 (2013).
74. Huang, Y.-L., Dobretsov, S., Ki, J.-S., Yang, L.-H. & Qian, P.-Y. Presence of acyl-homoserine lactone in subtidal biofilm and the implication in larval behavioral response in the polychaete *Hydroides elegans*. *Microb. Ecol.* **54**, 384–392 (2007).
75. Sneed, J. M., Sharp, K. H., Ritchie, K. B. & Paul, V. J. The chemical cue tetrabromopyrrole from a biofilm bacterium induces settlement of multiple Caribbean corals. *Proc. Biol. Sci.* **281**, 2013086 (2014).
76. Swanson, R. L. et al. Dissolved histamine: a potential habitat marker promoting settlement and metamorphosis in sea urchin larvae. *Mar. Biol.* **159**, 915–925 (2012).
77. Shikuma, N. J. et al. Marine tubeworm metamorphosis induced by arrays of bacterial phage tail-like structures. *Science* **343**, 529–533 (2014).
This work provides direct evidence that marine bacteria produce arrays of phage tail-like structures that trigger metamorphosis (an important stage of macrofouling) of the tubeworm *H. elegans*.
78. Shikuma, N. J., Antoshechkin, I., Medeiros, J. M., Pilhofer, M. & Newman, D. K. Stepwise metamorphosis of the tubeworm *Hydroides elegans* is mediated by a bacterial inducer and MAPK signaling. *Proc. Natl Acad. Sci. USA* **113**, 10097–10102 (2016).
79. Chatterjee, S. N. & Das, J. Electron microscopic observations on the excretion of cell-wall material by *Vibrio cholerae*. *J. Gen. Microbiol.* **49**, 1–11 (1967).
80. Kaparakis-Liaskos, M. & Ferrero, R. L. Immune modulation by bacterial outer membrane vesicles. *Nat. Rev. Immunol.* **15**, 375–387 (2015).
81. Freckelton, M. L., Nedved, B. T. & Hadfield, M. G. Induction of invertebrate larval settlement; different bacteria, different mechanisms? *Sci. Rep.* **7**, 42557 (2017).
82. Freckelton, M. L. et al. Bacterial lipopolysaccharide induces settlement and metamorphosis in a marine larva. *Proc. Natl Acad. Sci. USA* **119**, e2200795119 (2022).
83. Flemming, H.-C. et al. Biofilms: an emergent form of bacterial life. *Nat. Rev. Microbiol.* **14**, 563–575 (2016).
84. Maki, J. & Mitchell, R. Involvement of lectins in the settlement and metamorphosis of marine invertebrate larvae. *Bull. Mar. Sci.* **37**, 675–683 (1985).
85. Liang, X. et al. Bacterial cellulose synthesis gene regulates cellular c-di-GMP that control biofilm formation and mussel larval settlement. *Int. Biodeterior. Biodegrad.* **165**, 105330 (2021).
86. Peng, L.-H. et al. A bacterial polysaccharide biosynthesis-related gene inversely regulates larval settlement and metamorphosis of *Mytilus coruscus*. *Biofouling* **36**, 753–765 (2020).
87. Peng, L. H., Liang, X., Xu, J. K., Dobretsov, S. & Yang, J. L. Monospecific biofilms of *Pseudoalteromonas* promote larval settlement and metamorphosis of *Mytilus coruscus*. *Sci. Rep.* **10**, 2577 (2020).
88. Liang, X. et al. The flagellar gene regulates biofilm formation and mussel larval settlement and metamorphosis. *Int. J. Mol. Sci.* **21**, 710 (2020).
89. Denise, R., Abby, S. S. & Rocha, E. P. C. The evolution of protein secretion systems by co-option and tinkering of cellular machineries. *Trends Microbiol.* **28**, 372–386 (2020).
90. Maki, J. S., Ding, L., Stokes, J., Kavouras, J. H. & Rittschof, D. Substratum/bacterial interactions and larval attachment: films and exopolysaccharides of *Halomonas marina* (ATCC 25374) and their effect on barnacle cyprid larvae, *Balanus amphitrite* Darwin. *Biofouling* **16**, 159–170 (2000).
91. Maki, J. S., Rittschof, D. & Mitchell, R. Inhibition of larval barnacle attachment to bacterial films: an investigation of physical properties. *Microb. Ecol.* **23**, 97–106 (1992).
92. Maki, J. S., Rittschof, D., Costlow, J. D. & Mitchell, R. Inhibition of attachment of larval barnacles, *Balanus amphitrite*, by bacterial surface films. *Mar. Biol.* **97**, 199–206 (1988).
This work provides the first evidence that bacterial biofilms inhibit larval barnacle attachment and indicates that chemical inhibitors released from biofilms could be a root.
93. Alfaro, A. C., Young, T. & Ganesan, A. M. Regulatory effects of mussel (*Aulacomya maoriana* Iredale 1915) larval settlement by neuroactive compounds, amino acids and bacterial biofilms. *Aquaculture* **322–323**, 158–168 (2011).
94. Kon-ya, K. et al. Inhibitory effect of bacterial ubiquinones on the settling of barnacle, *Balanus amphitrite*. *Experientia* **51**, 153–155 (1995).
95. Holmström, C., James, S., Neilan, B. A., White, D. C. & Kjelleberg, S. *Pseudoalteromonas tunicata* sp. nov., a bacterium that produces antifouling agents. *Int. J. Syst. Evol. Microbiol.* **48**, 1205–1212 (1998).
96. Zeng, Z. et al. Pyromelanin from *Pseudoalteromonas lipolytica* reduces biofouling. *Microb. Biotechnol.* **10**, 1718–1731 (2017).

97. Matsumura, K., Nagano, M. & Fusetani, N. Purification of a larval settlement-inducing protein complex (SIPC) of the barnacle, *Balanus amphitrite*. *J. Exp. Zool.* **281**, 12–20 (1998).
98. Rittschof, D. & Cohen, J. H. Crustacean peptide and peptide-like pheromones and kairomones. *Peptides* **25**, 1503–1516 (2004).
99. Obeng, N. et al. Bacterial c-di-GMP has a key role in establishing host–microbe symbiosis. *Nat. Microbiol.* **8**, 1809–1819 (2023).
This work revealed that c-di-GMP underlies adaptation to a variety of hosts, from plants to humans, indicating that c-di-GMP is fundamental for establishing host association.
100. Paluch, E., Rewak-Soroczyńska, J., Jędrusik, I., Mazurkiewicz, E. & Jermakow, K. Prevention of biofilm formation by quorum quenching. *Appl. Microbiol. Biotechnol.* **104**, 1871–1881 (2020).
101. Liu, X., Yang, J.-L., Rittschof, D., Maki, J. S. & Gu, J.-D. Redirecting marine antibiofouling innovations from sustainable horizons. *Trends Ecol. Evol.* **37**, 469–472 (2022).
102. de la Fuente-Núñez, C., Reffuveille, F., Haney, E. F., Straus, S. K. & Hancock, R. E. W. Broad-spectrum anti-biofilm peptide that targets a cellular stress response. *PLoS Pathog.* **10**, e1004152 (2014).
103. Jiang, T., Qi, L., Hou, C., Fang, S. & Qin, W. Self-sterilizing polymeric membrane sensors based on 6-chloroindole release for prevention of marine biofouling. *Anal. Chem.* **92**, 12132–12136 (2020).
104. Feng, K., Ni, C., Yu, L., Zhou, W. & Li, X. Synthesis and antifouling evaluation of indole derivatives. *Ecotoxicol. Environ. Saf.* **182**, 109423 (2019).
105. Xiang, W. et al. Antifouling activity towards mussel by small-molecule compounds from a strain of *Vibrio alginolyticus* bacterium associated with sea anemone *Haliplanella* sp. *J. Microbiol. Biotechnol.* **27**, 460–470 (2017).
106. Wong, G. C. L. et al. Roadmap on emerging concepts in the physical biology of bacterial biofilms: from surface sensing to community formation. *Phys. Biol.* **18**, 051501 (2021).
107. Zhao, Y. et al. Pulsed hydraulic-pressure-responsive self-cleaning membrane. *Nature* **608**, 69–73 (2022).
108. Bhattacharjee, A., Khan, M., Kleiman, M. & Hochbaum, A. I. Effects of growth surface topography on bacterial signaling in coculture biofilms. *ACS Appl. Mater. Interfaces* **9**, 18531–18539 (2017).

Acknowledgements

This study was funded by the National Natural Science Foundation of China (grant nos. 32100101, 32370105, 92051103, 41876159 and 31970036), the Fundamental Research Funds for the Central Universities (grant no. 1225011021289), the Program of Shanghai Academic Research Leader (grant no. 20XD1421800) and the National Key Research and Development Program of China (grant nos. 2019YFC0312104 and 2021YFA0910302).

Author contributions

X.L., W.L. and J.-D.G. contributed to discussions and wrote the manuscript. L.Z. and B.L. prepared the figures and wrote the manuscript. P.D.M., J.-L.Y., D.R. and J.M. reviewed the literature and edited the contents of macro-biofouling.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Xiaobo Liu, Weijie Liu or Ji-Dong Gu.

Peer review information *Nature Chemical Biology* thanks Manfred Zinn, Xiaohui Zhou and the other, anonymous reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© Springer Nature America, Inc. 2024